

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0585

Zeeland Generating Station 138 kV
June 27, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0585. The Interconnection Customer proposes a 250.6 MW solar photovoltaic facility interconnecting at the Zeeland Generating Station 138 kV substation in the West Virginia area. The request was received on February 27, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,373,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BECK ROAD - SPORN 230 KV	INTERNATIONAL PAPER - FLATLICK SUBSTATION 500 KV	122.4	508	6.2
CEDARCLIFF - JEFFERSON RD 138 KV	MT. VIEW - ZEELAND GENERATING STATION 138 KV line	119.0	195	100.0
ZEELAND GENERATING STA - CEDAR CLIFF 138 KV	MT. VIEW - ZEELAND GENERATINGSTATION 138 KV Line	115.1	209	100.0
E PROCTORVILLE - JEFFERSON ROAD 138 KV	MT. VIEW - ZEELAND GENERATING STATION 138 KV Line	113.5	191	100.0
MT. VIEW - ZEELAND GENERATING STATION 138 KV	ZEELAND GENERATING STATION - CEDAR CLIFF 138 KV	108.0	223	100.0
JONES CHAPEL SUBSTATION - DEKALB SUBSTATION 500 KV	ROBISON PARK - RUSSELL FORK 230 KV	100.4	2,860	11.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BECK ROAD - SPORN 230 KV	Rebuild line with higher-capacity conductor	\$6,421,000	6% (\$370,000)
CEDARCLIFF - JEFFERSON RD 138 KV	Rebuild line with higher-capacity conductor	\$8,561,000	100% (\$8,561,000)
ZEELAND GENERATING STA - CEDAR CLIFF 138 KV	Rebuild line with higher-capacity conductor	\$3,135,000	100% (\$3,135,000)
E PROCTORVILLE - JEFFERSON ROAD SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$1,100,000	100% (\$1,100,000)
ZEELAND GENERATING STA - MT. VIEW 138 KV line	Reconductor line	\$1,444,000	100% (\$1,444,000)
DEKALB - JONES CHAPEL 500 KV	Reconductor line	\$69,255,000	2% (\$1,763,000)
	Total allocated to project		\$16,373,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.